

RepreGuard: Detecting LLM-Generated Text by Revealing Hidden Representation Patterns

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Abstract

Detecting content generated by large language models (LLMs) is crucial for preventing misuse and building trustworthy AI systems. Although existing detection methods perform well, their robustness in out-of-distribution (OOD) scenarios is still lacking. In this paper, we hypothesize that, compared to features used by existing detection methods, the internal representations of LLMs contain more comprehensive and raw features that can more effectively capture and distinguish the statistical pattern differences between LLM-generated texts (LGT) and human-written texts (HWT). We validated this hypothesis across different LLMs and observed significant differences in neural activation patterns when processing these two types of texts. Based on this, we propose **RepreGuard**, an efficient statistics-based detection method. Specifically, we first employ a surrogate model to collect representation of LGT and HWT, and extract the distinct activation feature that can better identify LGT. We can classify the text by calculating the projection score of the text representations along this feature direction and comparing with a precomputed threshold. Experimental results show that RepreGuard outperforms all baselines with average 94.92% AUROC on both in-distribution (ID) and OOD scenarios, while also demonstrating robust resilience to various text sizes and mainstream attacks.¹

1 Introduction

LLMs demonstrate impressive language understanding and generation capabilities (OpenAI, 2022; Anthropic, 2023; Ghahramani, 2023;

MetaAI, 2024), enabling them to produce creative and persuasive content that aligns with human preferences (Wu et al., 2023). These capabilities have raised concerns regarding future data regulation, particularly due to biases (Tjauatja et al., 2023) and hallucinations (Ji et al., 2023) in LGT. Moreover, the potential misuse of LLMs, such as generating fake news (Pagnoni et al., 2022) or facilitating academic dishonesty (Cotton et al., 2024), poses significant risks and challenges to society. To defend against these usage cases, several algorithms have been developed to detect LGT. These primarily include fine-tuning-based classifiers, which involve training an model with extensive labeled data for classification, such as the OpenAI detector (Solaiman et al., 2019), and statistics-based methods, which identify LGT using feature metrics from specific distributions in a small training dataset, such as DetectGPT (Mitchell et al., 2023).

Fine-tuning-based classifiers generally offer higher accuracy than statistics-based detectors, but require large amounts of labeled data and often struggle to generalize across different generators, making updates costly for new models (Guo et al., 2023). In contrast, statistics-based methods provide better interpretability and only require setting a threshold based on observed distribution patterns in a small sample size, offering stonger reliability for real-world applications (Wu et al., 2023). However, current statistics-based methods perform poorly in both ID and OOD scenarios due to the insufficient robustness in classification feature metrics. For example, varying prompts could control the perplexity of generated text, rendering thresholds from training samples ineffective (Hans et al., 2024). This limitation is exacerbated in OOD scenarios, posing significant challenges to the usability of statistics-based detectors, with the growing number of new LLMs.

In response to this challenge, we propose a de-

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¹Data and code are publicly available at: <https://github.com/NLP2CT/RepreGuard>

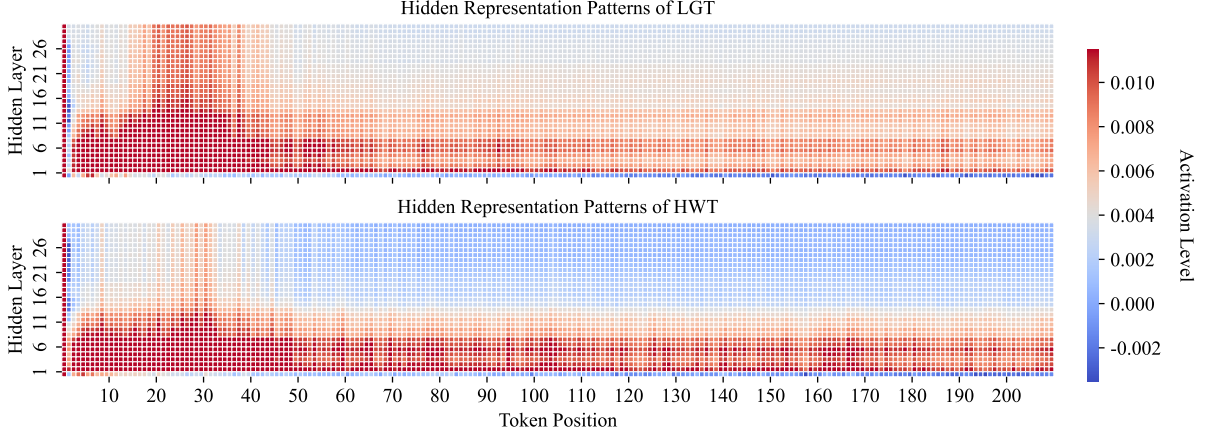


Figure 1: Comparison of Average Hidden Representation Distribution in Llama-3.1-8B Using 1000 Pairs of LGT and HWT from 4 Different LLMs. The red represents active representations and blue represents relatively inactive representations. The depth of the color represents the level of representation activation.

tection method based on hidden representation to identify texts generated by LLMs. This detection method is motivated by the following hypothesis: LLMs exhibit distinct hidden representation patterns when processing LGT compared to HWT, due to their different perceptions of statistical patterns in these text types.

These hidden representation patterns differences can be more explicitly observed in the model’s representations, which are higher-dimensional features. This is also where the detection abilities of existing statistics-based detectors come into play, utilizing classification feature metrics such as likelihood, rank, and other variants. Thus, the model’s representations may encompass more comprehensive and raw features that can enhance the ability to distinguish between text types, with a stronger potential for identifying LGT. To achieve this, we first employ a surrogate model as an “observer” to obtain the representations when processing texts generated by LLMs and those written by humans. We observed that the hidden representation patterns of the two types of texts show distinct differences, which serve as a strong signal for detecting LGT. Then, to filter out noisy features that might hinder LGT detection, we perform dimensionality reduction and modeling on both types of text representations to identify features that maximally retain the distinguishing information. By calculating the projection score of the representation of a given text along this feature direction, which we termed “RepreScore”, and comparing it to the optimal classification threshold statistically derived,

we can determine whether the text was generated by an LLM or written by a human. We call this detection method **RepreGuard**, which is an efficient detection method that combines the strengths of both statistics-based and fine-tuning-based approaches. RepreGuard exhibits zero-shot characteristics, requiring only a small number of training samples from one LLM source to generalize across texts generated by various types of LLMs.

Experiments on different setups show that RepreGuard outperforms the SOTA RoBERTa-based classifier and the statistics-based method Binoculars in both ID and OOD scenarios, with an average of 11.05% and 05.88% AUROC higher than RoBERTa-based classifier and Binoculars, respectively, and achieve better performance with fewer training samples. In addition, our method demonstrates strong robustness to the generalization on Domains, texts with varied sizes, mainstream attacks including text paraphrasing and perturbation attacks and various sampling methods. It also achieves an excellent balance between effectiveness and resource efficiency.

2 Related Works

Statistics-based Detection Methods The statistics-based detection method detects LGT by examining the distribution difference of the specified feature metrics between LGT and HWT. This classification is achieved by extracting these feature metrics from the given text and comparing them to thresholds derived from statistics on training datasets, without the need to fine-tune a neural model as classifier. Early statistics-based

methods focused on calculating feature metrics derived from the model output logits, such as Entropy, Log-Likelihood, and Log-Rank (Solaiman et al., 2019). Subsequently, Su et al. (2023) introduced the Log-Likelihood Log-Rank Ratio (LRR), offering a more comprehensive evaluation by calculating the ratio of Log-Likelihood to Log-Rank. Recently, perturbation-based methods have gained significant attention. Mitchell et al. (2023) and Su et al. (2023) used Log-Likelihood and Log-Rank curvature, respectively, to identify LGT, based on the hypothesis that LGT maintains higher Log-Likelihood and Log-Rank after semantic perturbation compared to HWT. Bao et al. (2024) enhanced this by replacing the perturbation step in DetectGPT with a more efficient sampling procedure, reducing the computational cost. Other methods, like DNA-GPT (Yang et al., 2024), use an iterative process where an LLM extends truncated text and assesses authorship via probability differences between the original and extended text. In contrast, GECScore (Wu et al., 2025) relies on the finding that LLMs are more likely to correct grammatical errors in human-written text and distinguishes text sources by measuring changes in similarity before and after grammatical error correction. Binoculars (Hans et al., 2024) is a novel method that employs a pair of LLMs to calculate the ratio of perplexity and cross-perplexity, measuring how one model’s prediction of the next token surprises another one.

Fine-Tuning-Based Detection Methods Another approach is to fine-tune a neural model as a classifier for distinguishing between HWT and LGT, typically requiring a large amount of labeled data. Early efforts focused on fine-tuning pre-trained classifiers for detecting news articles (Zellers et al., 2019) and social media content (Fagni et al., 2020). Recent studies (Guo et al., 2023; Liu et al., 2023; Chen et al., 2023; Wang et al., 2023) further confirmed the strong performance of fine-tuned language model in identifying LGT. OpenAI’s detector, for example, is a fine-tuned RoBERTa-based classifier to perform this task (Solaiman et al., 2019). However, fine-tuning-based classifiers tend to overfit to their training data or the training distribution of the source model, resulting in performance drops when encountering new LLMs or domains data (Sarvazyan et al., 2023).

3 RepreGuard

3.1 Preliminary

Hypothesis *The premise of RepreGuard is that LLMs perceive the statistical patterns of LGT and HWT differently. Their hidden representation patterns exhibit noticeable differences when observing and processing these two types of text.*

This hypothesis arises from the idea that there are behavioral differences in the writing processes between LGT and HWT, which can be captured through internal hidden representations, as the internal hidden representations of LLMs typically contain more information and raw features compared to external behaviours. This information can reveal the model’s intrinsic understanding of the differences between the LGT and HWT.

Internal Representation of LLMs Recent research (Voita et al., 2024; Durrani et al., 2020; Xu et al., 2024b) suggest that the internal representation mechanisms of LLMs may capture more information. For instance, LLM-Check (Sriraman et al., 2024) extracts the hidden representations from each layer during response generation, and calculates their covariance matrices (Hidden Score) as an indicator for hallucination detection. Zhang et al. (2024) indirectly reveals the spatial reasoning capabilities encoded in the hidden representations of vision-language models by systematically analyzing output probabilities under continuous variations in input space. Tang et al. (2024) proposed a method of extracting latent representations as specific concept directions, thereby enabling the effective guidance and control of LLMs towards the concept direction (eg, safety and honesty).

Therefore, RepreGuard is designed to capture the underlying hidden representations that characterise these processes based on the distinct behavioral processes in human writing and AI writing. These distinctions likely result from the statistical patterns internalized by LLMs during training and how these are leveraged in generative tasks. To validate this hypothesis, we follow Zou et al. (2023) and compare the neural activation patterns in Llama-3.1-8B when processing 1000 pairs of LGT and HWT from 4 different LLMs, as illustrated in Figure 1. Specifically, LGT and HWT exhibit largely similar hidden representations during the early stage of the sequence (approximately tokens 0–20). However, when the number of tokens